

CUSTOMER RETENTION & CHURN ANALYSIS

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INTERNSHIP: Data Science & Analytics

TOOLS USED:

- PYTHON
- PANDAS
- MATPLOTLIB
- GOOGLE COLAB

1.INTRODUCTION:

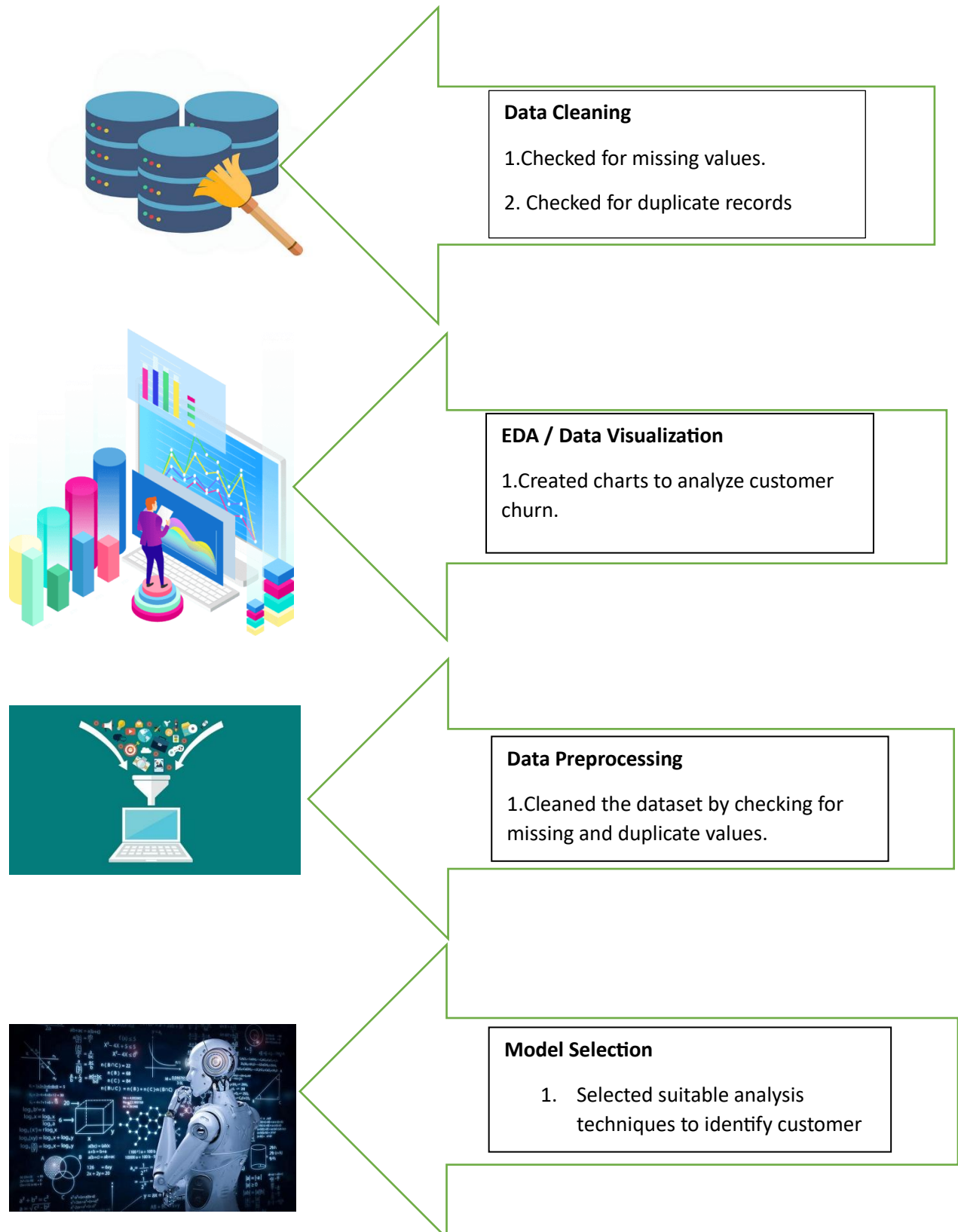
Customer retention is a critical factor for the success of any business. Retaining existing customers is often more cost-effective than acquiring new ones.

Customer churn occurs when customers stop using a company's products or services, which can negatively impact revenue and business growth.

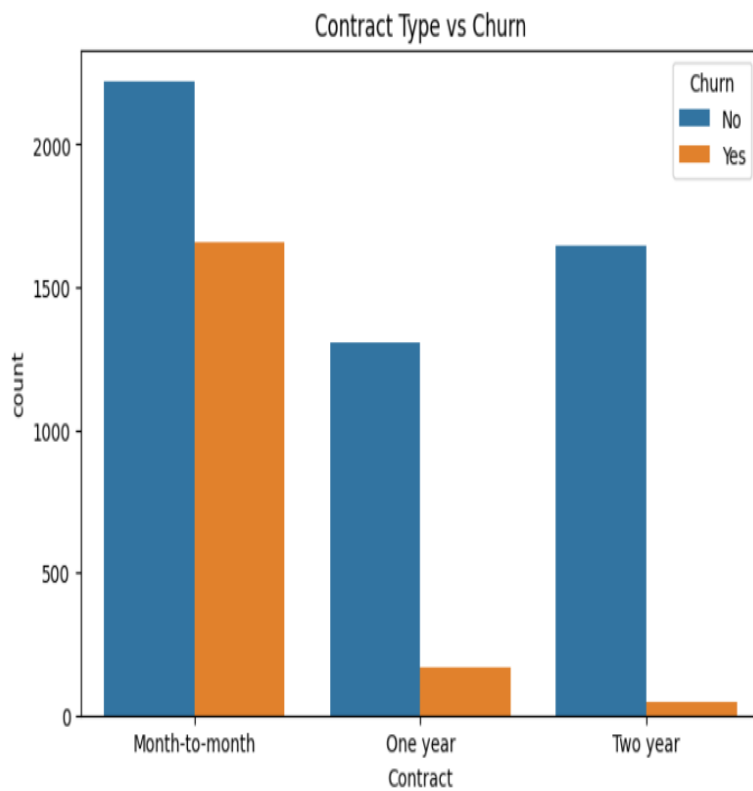
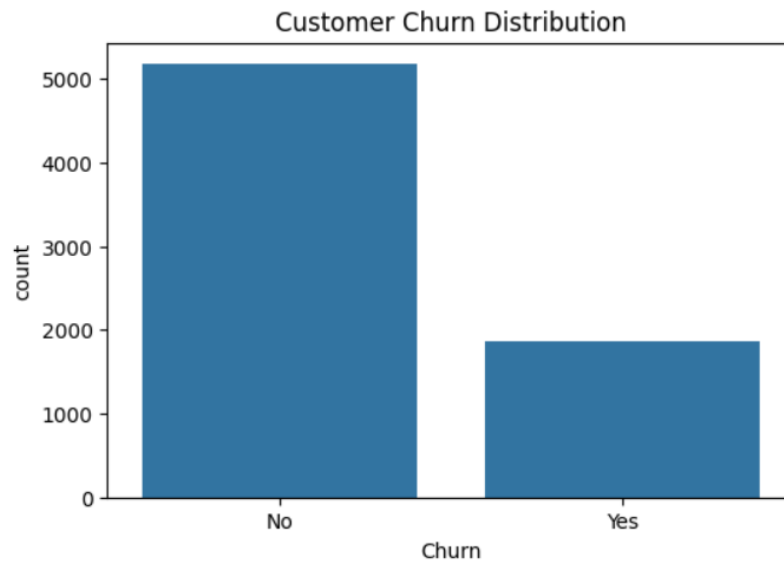
This project focuses on Customer Retention and Churn Analysis using customer data. The objective is to identify patterns and factors that contribute to customer churn, analyze customer behavior, and gain insights into customer retention trends. Various attributes such as gender, tenure, contract type, payment method, internet service, and monthly charges are analyzed to understand their impact on customer churn.

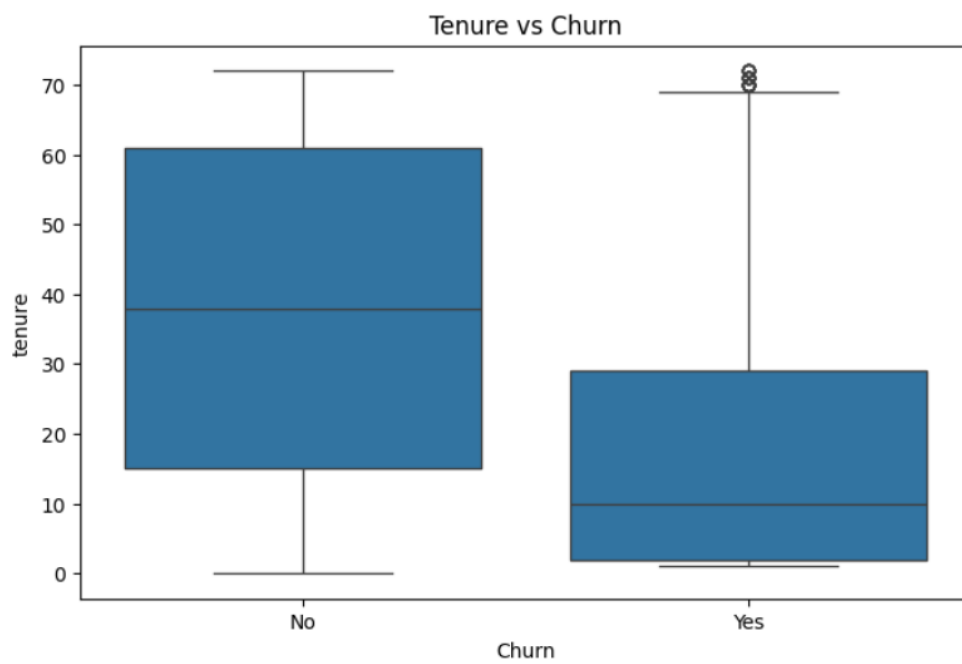
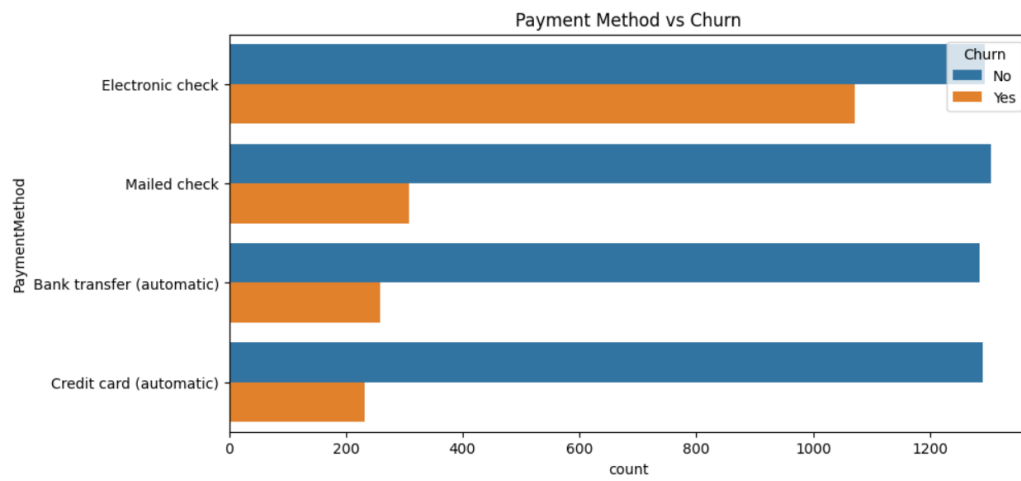
By applying data analysis and visualization techniques, this project helps businesses identify high-risk customers and develop effective retention strategies. The findings can support data-driven decision-making, improve customer satisfaction, and enhance long-term business performance.

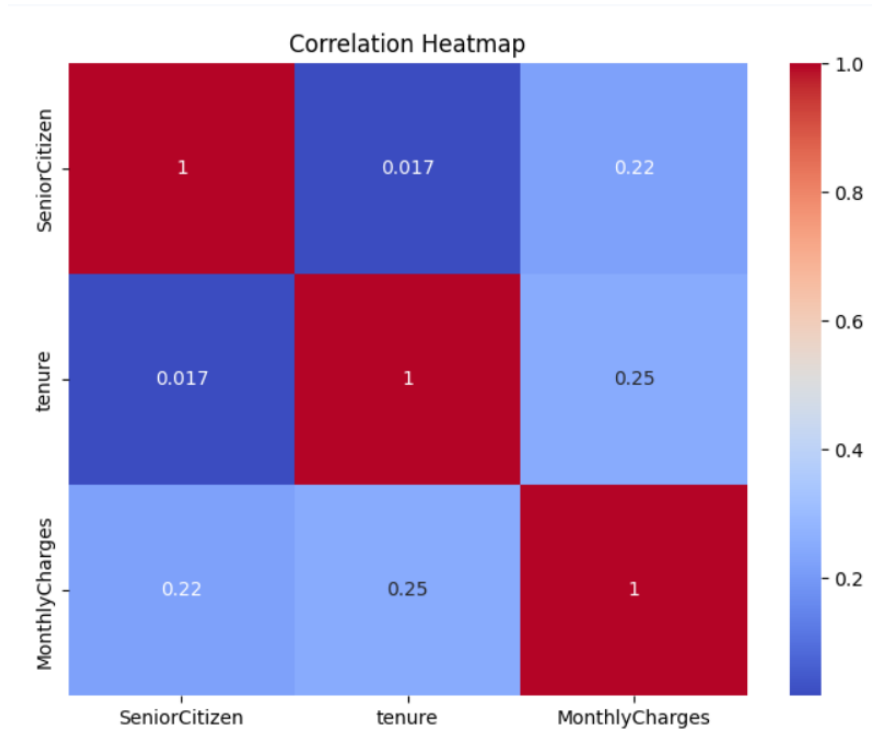
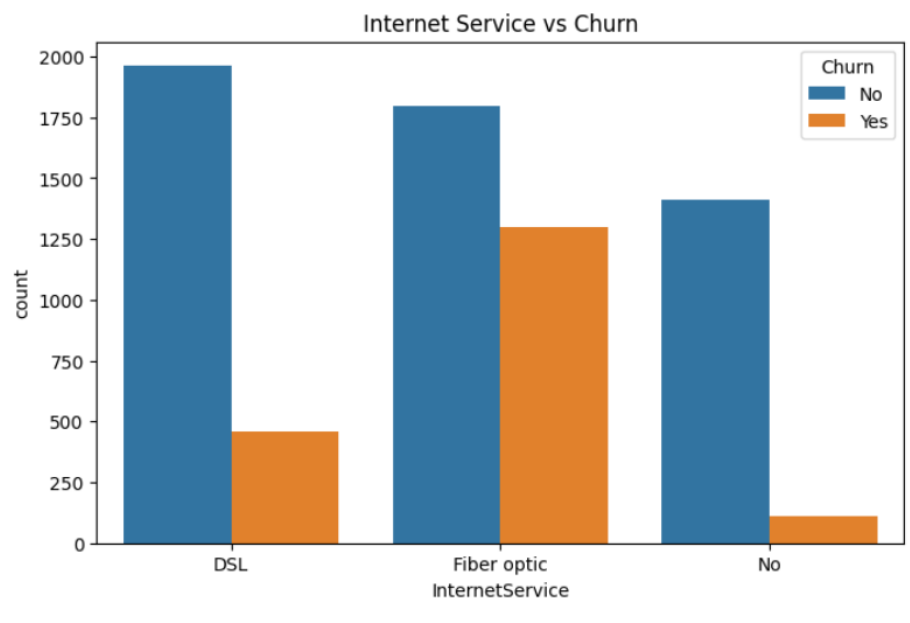
Project Architecture / Project Flow



ANALYSIS & CHARTS:







Key Insights

1. Customers with month-to-month contracts exhibited the highest churn rate compared to customers with long-term contracts.
2. Customers with shorter tenure were more likely to discontinue the service, indicating that new customers are at a higher risk of churn.
3. Electronic check users showed a higher churn rate than customers using other payment methods.
4. Customers with higher monthly charges demonstrated a greater tendency to leave the service.
5. Long-term customers displayed stronger loyalty and better retention rates.
6. Contract type, payment method, tenure, and monthly charges were identified as the major factors influencing customer churn.

Recommendations

1. Encourage customers to switch to long-term contracts by offering discounts and special benefits.
2. Improve customer engagement during the initial months of service to reduce early customer churn.
3. Provide personalized offers and loyalty rewards to high-risk customers.
4. Enhance customer support services to improve customer satisfaction and retention.
5. Develop targeted retention campaigns for customers with high monthly charges.
6. Continuously monitor customer behavior and implement proactive retention strategies.

Conclusion:

This project successfully analyzed customer retention and churn patterns using data analytics techniques. Various factors such as contract type, tenure, payment method, internet service, and monthly charges were examined to understand their impact on customer churn.

The analysis revealed that customers with month-to-month contracts, shorter tenure, and higher monthly charges were more likely to leave the service. Through data visualization and exploratory analysis, valuable insights were obtained regarding customer behavior and retention trends.

The findings of this project can help organizations develop effective customer retention strategies, improve customer satisfaction, reduce churn rates, and support data-driven decision-making. Overall, the project demonstrates the importance of data analytics in understanding customer behavior and enhancing long-term business performance.